

3.2 Linear Momentum & Conservation

Question Paper

Course	CIEA Level Physics
Section	3. Dynamics
Topic	3.2 Linear Momentum & Conservation
Difficulty	Medium

Time allowed: 80

Score: /60

Percentage: /100

Question 1a

A ball X of mass 140 g travelling at constant velocity v collides head-on with a stationary ball Y of mass 620 g, as shown in Fig. 1.1.

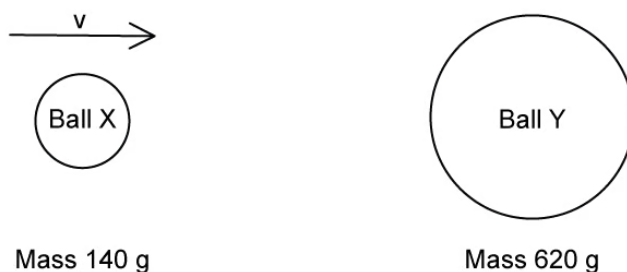


Fig. 1.1

After the collision, ball X comes to a stop. Ball Y moves off in the same initial direction as X with a velocity of 0.8 m s^{-1} .

Determine the velocity v of ball X before the collision.

[3 marks]

Question 1b

(i)

Show that the collision in (a) is inelastic.

[3]

(ii)

Explain why the collision in (a) is inelastic.

[1]

[4 marks]

Question 1c

The balls are replaced by an aluminium sphere of mass 2.7 kg and a steel sphere of 7.9 kg as shown in Fig. 1.2.

The aluminium sphere has an initial velocity of 10.0 m s^{-1} when it collides with the stationary steel sphere. Immediately after the collision, the velocity of the steel sphere is 5.1 m s^{-1} .

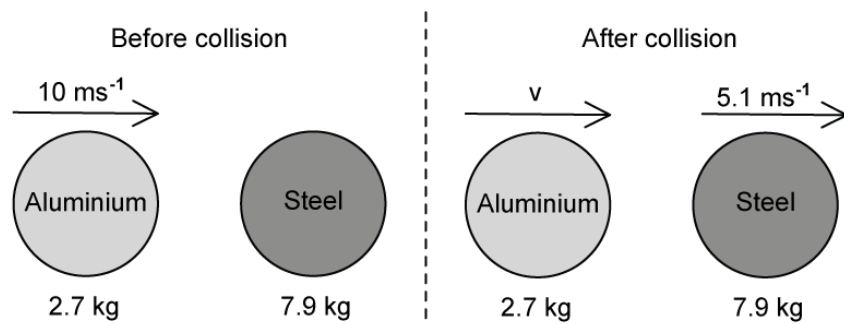


Fig. 1.2

Calculate the velocity v of the aluminium sphere immediately after the collision.

[3 marks]

Question 1d

Verify that the collision in (c) is elastic.

[3 marks]

Question 2a

Two identical blocks A and B each of mass 400 g are travelling towards each other along a straight line through their centre, as shown in Fig. 1.1.



Fig. 1.1

Both blocks are moving at a speed of 0.36 m s^{-1} relative to the surface.

As a result of the collision, the blocks reverse their direction of motion and travel at the same speed as each other. During the collision, 33% of the kinetic energy of the blocks is transferred to the surroundings as thermal energy.

(i)

State the total momentum of the blocks before the collision.

[1]

(ii)

State and explain whether the collision is elastic or inelastic.

[2]

[3 marks]

Question 2b

Calculate the final speed of the blocks.

[3 marks]

Question 2c

The duration of the collision between the blocks is 750 ms.

Determine the average force exerted by one block on the other.

[3 marks]

Question 2d

The same blocks are investigated with different initial speeds. Block A is launched towards B at a speed of 0.6 m s^{-1} and B is launched towards A at a speed of 0.4 m s^{-1} .

Once the blocks collide, they stick together and move as one after the collision.

Determine the speed of the combined blocks after the collision and state the direction.

[4 marks]

Question 3a

A fast-moving asteroid B collides with a slower-moving asteroid A as shown in Fig. 1.1. Asteroid B becomes embedded within asteroid A as shown.

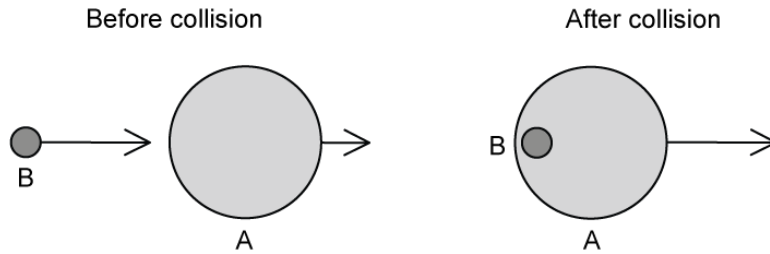


Fig. 1.1

Before the collision, asteroid A had a velocity of 3.61 km s^{-1} and a momentum of $2.00 \times 10^{17} \text{ kg m s}^{-1}$.

(i)

State the principle of conservation of linear momentum.

[2]

(ii)

Show that the mass of asteroid A was about $5.5 \times 10^{13} \text{ kg}$.

[2]

[4 marks]

Question 3b

Calculate the combined velocity, in km s^{-1} , of the asteroids after the collision.

mass of asteroid B = $6.40 \times 10^{12} \text{ kg}$

velocity of asteroid B before the collision = 14.0 km s^{-1}

[3 marks]

Question 3c

A short time later, the combined asteroid AB collides with a stationary asteroid C. Immediately following the collision, the asteroids are deflected by the angles shown in Fig. 1.2.

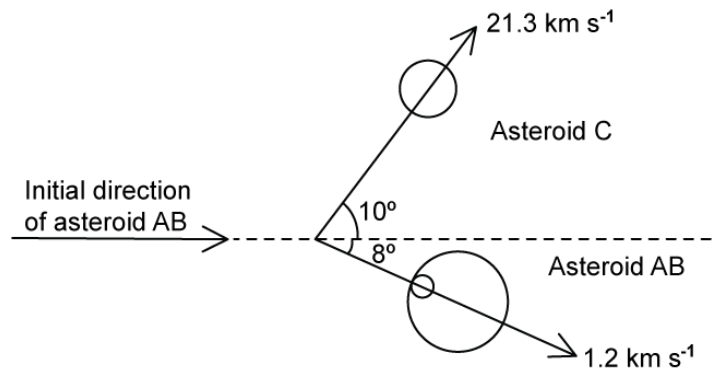


Fig. 1.2

Show that the mass of asteroid C is about $1.0 \times 10^{13} \text{ kg}$.

[4 marks]

Question 3d

Determine if the collision in (c) is elastic or inelastic.

[3 marks]

Question 4a

Ball X has a mass of 50 g and is supported by a long string, as shown in Fig. 1.1.

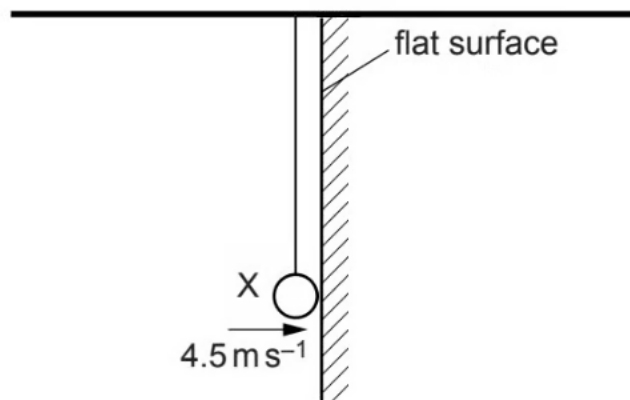


Fig. 1.1

The ball is pulled back and released. The ball collides normally with a flat surface at a velocity of 4.5 m s^{-1} and rebounds elastically.

The positive direction is horizontal and to the right.

Determine the total change in momentum of ball X.

[2 marks]

Question 4b

Ball X is hung next to ball Y which is supported by a string of the same length, as shown in Fig. 1.2.

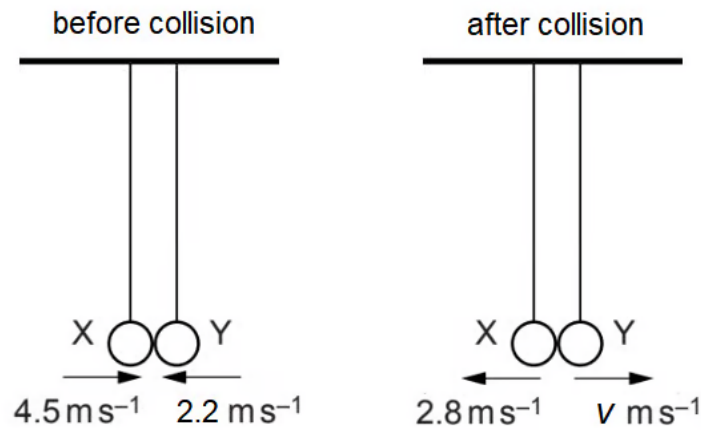


Fig. 1.2

Ball Y has a mass of 125 g.

The balls are each pulled back and pushed towards each other. When the balls collide, the strings are vertical. The balls rebound in opposite directions.

The velocities of X and Y just before and just after the collision are shown in Fig. 1.2.

Use the conservation of linear momentum to determine the rebound velocity v of Y.

[3 marks]

Question 4c

Deduce whether the collision, in **part (b)**, is elastic or inelastic.

[2 marks]

Question 4d

Use Newton's laws to explain why the magnitude of the change in momentum of each ball is the same.

[3 marks]

Question 5a

A shell of mass 30 g is fired from the barrel of a rifle of mass 1.9 kg. Immediately after being fired, the bullet has a momentum of 36 kg m s^{-1} .

State the total momentum of the rifle and the bullet before the rifle is fired and give a reason for your answer.

[2 marks]

Question 5b

Calculate the velocity of the shell just after the rifle is fired.

[2 marks]

Question 5c

(i)

Using the principle of conservation of momentum, and your answer to part (a), determine the total momentum of the rifle and the shell.

[1]

(ii)

Calculate the recoil momentum of the rifle.

[2]

[3 marks]**Question 5d**

The shell has a momentum of 9.8 kg m s^{-1} just before it hits a target.

It takes 4.5 ms for the bullet to be stopped by the target.

Calculate the average force needed to stop the bullet.

[3 marks]

Model Answers: Medium

1a

(a) Determine the velocity v of ball X before the collision:

List the known quantities:

- Mass of X = 140 g
- Mass of Y = 620 g
- Velocity of X before the collision = v
- Velocity of X after the collision = 0
- Velocity of Y before the collision = 0
- Velocity of Y after the collision = 0.8 m s^{-1}

Use conservation of momentum to write an expression for v :

- Momentum before collision: $m_X v_X + m_Y v_Y = 140v + 0$
- Momentum after collision: $m_X v_X + m_Y v_Y = 0 + (620 \times 0.8)$
- Conservation of momentum: Momentum before collision = Momentum after collision
- Conservation of momentum: $140v = 620 \times 0.8$ [1 mark]
- Rearrange: $v = \frac{620 \times 0.8}{140}$ [1 mark]
- $v = 3.543 = 3.5 \text{ m s}^{-1}$ [1 mark]

[Total: 3 marks]

1b

(b)

(i) Show that the collision in **(a)** is inelastic:

List the known quantities:

- Mass of X = 140 g = 0.14 kg
- Mass of Y = 620 g = 0.62 kg
- Velocity of X before the collision = 3.5 m s^{-1}
- Velocity of X after the collision = 0
- Velocity of Y before the collision = 0
- Velocity of Y after the collision = 0.8 m s^{-1}

Determine the kinetic energy before and after the collision:

- KE before = $\frac{1}{2}m_X v_X^2 + \frac{1}{2}m_Y v_Y^2 = \left(\frac{1}{2} \times 0.14 \times 3.5^2\right) + 0$
- KE before = 0.8575 = 0.86 J [1 mark]
- KE after = $\frac{1}{2}m_X v_X^2 + \frac{1}{2}m_Y v_Y^2 = 0 + \left(\frac{1}{2} \times 0.62 \times 0.8^2\right)$
- KE after = 0.1984 = 0.20 J [1 mark]

Write a concluding sentence:

- The kinetic energy after the collision is less than the kinetic energy before the collision, hence it is an inelastic collision; [1 mark]

(ii) The collision is inelastic because:

- Some of X's (kinetic) energy has been transferred to other forms of energy (such as heat and sound); [1 mark]

[Total: 4 marks]

1c

(c) Calculate the velocity of the aluminium sphere immediately after the collision:

List the known quantities:

- Mass of aluminium sphere, $m_{Al} = 2.7 \text{ kg}$
- Mass of steel sphere, $m_S = 7.9 \text{ kg}$
- Initial velocity of aluminium sphere, $u_{Al} = 10.0 \text{ m s}^{-1}$
- Initial velocity of steel sphere, $u_S = 0$
- Final velocity of aluminium sphere = v
- Final velocity of steel sphere, $v_S = 5.1 \text{ m s}^{-1}$

Use conservation of momentum to write an expression for v :

- Momentum before collision: $m_{Al}u_{Al} + m_Su_S = (2.7 \times 10) + 0$
- Momentum after collision: $m_{Al}v_{Al} + m_Sv_S = (2.7 \times v) + (7.9 \times 5.1)$
- Conservation of momentum: Momentum before collision = Momentum after collision
- $(2.7 \times 10) = 2.7v + (7.9 \times 5.1) \Rightarrow v = \frac{27 - (7.9 \times 5.1)}{2.7}$ [1 mark]
- $v = -4.922 = -4.9 \text{ m s}^{-1}$ [1 mark]

[Total: 3 marks]

- The negative sign is necessary for the final answer
 - This tells us that the aluminium sphere **changes** direction
- Remember to show your working – the first mark is for **substitution**, not just stating the equation!

1d

(d) Verify that the collision in **(c)** is elastic:

List the known quantities:

- Mass of aluminium sphere, $m_{Al} = 2.7 \text{ kg}$
- Mass of steel sphere, $m_S = 7.9 \text{ kg}$
- Initial velocity of aluminium sphere, $u_{Al} = 10.0 \text{ m s}^{-1}$
- Initial velocity of steel sphere, $u_S = 0$

- Final velocity of aluminium sphere, $v_{Al} = 4.9 \text{ m s}^{-1}$
- Final velocity of steel sphere, $v_S = 5.1 \text{ m s}^{-1}$

Determine the kinetic energy before and after the collision:

- KE before = $\frac{1}{2} m_{Al} u_{Al}^2 + \frac{1}{2} m_S u_S^2 = \left(\frac{1}{2} \times 2.7 \times 10^2 \right) + 0$
- KE before = 135 J [1 mark]
- KE after = $\frac{1}{2} m_{Al} v_{Al}^2 + \frac{1}{2} m_S v_S^2 = \left(\frac{1}{2} \times 2.7 \times 4.9^2 \right) + \left(\frac{1}{2} \times 7.9 \times 5.1^2 \right)$
- KE after = 135.15 = 135 J [1 mark]

Write a concluding sentence:

- The kinetic energy after the collision is the same as the kinetic energy before the collision, hence it is an elastic collision; [1 mark]

[Total: 3 marks]

2a

(a)

(i) The total momentum of the blocks before the collision is:

- Total momentum = $mu + mv = (400 \times 0.36) + (400 \times -0.36)$
- Total momentum = 0 [1 mark]

(ii) The collision is:

- Inelastic; [1 mark]

Because:

- 33% of the kinetic energy is lost as heat **OR** in elastic collisions, 100% of kinetic energy is conserved; [1 mark]

[Total: 3 marks]

Most scenarios where two objects collide and then move in opposite directions are commonly elastic. However, there is information given in the question that suggests that this isn't the case for this scenario.

Always read the full question carefully, and don't just assume one or the other!

2b

(b) Calculate the final speed of the blocks:

List the known quantities:

- Energy lost = 33% = 0.33 $\Rightarrow E_{k(2)} = 0.66 \times E_{k(1)}$
- Initial speed of each block, $u = 0.36 \text{ m s}^{-1}$
- Final speed of each block = v

Write an expression for the ratio of the initial and final kinetic energies:

- Kinetic energy: $E_k = \frac{1}{2}mv^2 \Rightarrow E_k \propto v^2$
- Initial kinetic energy: $E_{k(1)} \propto u^2$
- Final kinetic energy: $E_{k(2)} \propto v^2$

(b) Calculate the final speed of the blocks:

List the known quantities:

- Energy lost = 33% = 0.33 $\Rightarrow E_{k(2)} = 0.66 \times E_{k(1)}$
- Initial speed of each block, $u = 0.36 \text{ m s}^{-1}$
- Final speed of each block = v

Write an expression for the ratio of the initial and final kinetic energies:

- Kinetic energy: $E_k = \frac{1}{2}mv^2 \Rightarrow E_k \propto v^2$
- Initial kinetic energy: $E_{k(1)} \propto u^2$
- Final kinetic energy: $E_{k(2)} \propto v^2$
- $\frac{E_{k(1)}}{E_{k(2)}} = \frac{u^2}{v^2}$ [1 mark]

Determine the final speed of the blocks, v :

- $\frac{E_{k(1)}}{0.66 \times E_{k(1)}} = \frac{0.36^2}{v^2} \Rightarrow v^2 = 0.36^2 \times 0.66$
- $v = 0.36 \times \sqrt{0.66}$ [1 mark]
- $v = 0.2924 = 0.29 \text{ m s}^{-1}$ [1 mark]

[Total: 3 marks]

2c

(c) Determine the average force exerted by one block on the other:

List the known quantities:

- Initial speed of both blocks, $u = 0.36 \text{ m s}^{-1}$

- Final speed of both blocks, $v = 0.29 \text{ m s}^{-1}$ (from **(b)**)
- Mass of each block, $m = 400 \text{ g} = 0.4 \text{ kg}$
- Time during the collision, $\Delta t = 750 \text{ ms} = 750 \times 10^{-3} \text{ s}$

Calculate the change in velocity, Δv

- $\Delta v = v - u = 0.36 - (-0.29) = 0.65 \text{ m s}^{-1}$ [1 mark]

Calculate the average force, F :

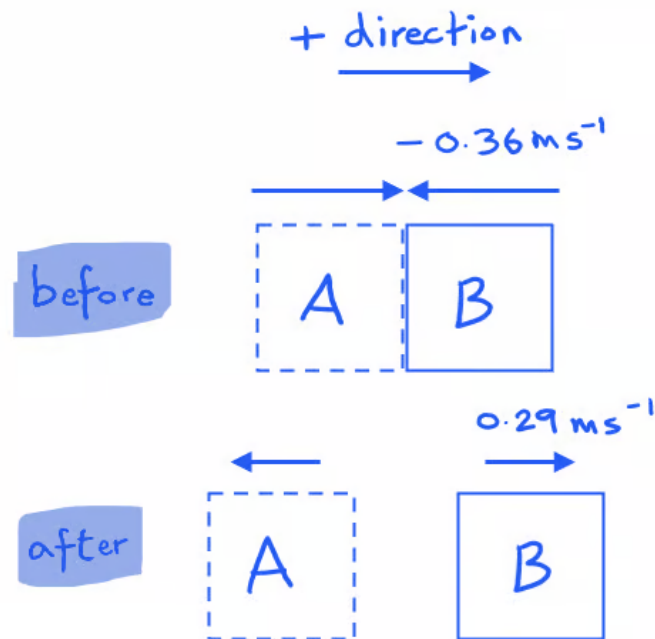
- Momentum: $p = mv$
- Force: $F = \frac{\Delta p}{\Delta t} = \frac{m \Delta v}{\Delta t}$
- $F = \frac{0.4 \times 0.65}{750 \times 10^{-3}}$ [1 mark]
- $F = 0.35 \text{ N}$ [1 mark]

[Total: 3 marks]

A common mistake in this question is calculating the change in velocity Δv as $0.36 - 0.29 = 0.07 \text{ m s}^{-1}$

This is incorrect as it is not treating velocity as a **vector** which has both magnitude and direction

Make sure to consider just one of the blocks, and be clear on which direction you are taking as the positive direction:



The Δv equation is always final velocity v – initial velocity u . Whether you take the right or left as positive, the magnitude of the force as the final answer will still be the same.

2d

(d) Determine the speed and direction of the combined blocks after the collision:

List the known quantities:

- Initial velocity of block A, $v_A = 0.6 \text{ ms}^{-1}$
- Initial velocity of block B, $v_B = -0.4 \text{ ms}^{-1}$
- Mass of block A, $m_A = 400 \text{ g} = 0.4 \text{ kg}$
- Mass of block B, $m_B = 400 \text{ g} = 0.4 \text{ kg}$

Use conservation of linear momentum:

- Conservation of momentum: Total momentum before the collision = Total momentum after the collision
- Conservation of momentum: $m_A v_A + m_B v_B = (m_A + m_B)v$ [1 mark]

• Rearrange: $v = \frac{m_A v_A + m_B v_B}{m_A + m_B} \Rightarrow v = \frac{m(v_A + v_B)}{2m}$

- $v = \frac{0.4(0.6 - 0.4)}{2 \times 0.4}$ [1 mark]
- $v = 0.1 \text{ m s}^{-1}$ [1 mark]

State the final speed and direction:

- Speed after the collision = 0.1 m s^{-1}
- Direction = same as initial direction of A; [1 mark]

[Total: 4 marks]

3a

(a)

(i) The principle of conservation of linear momentum states that:

- The momentum of a system remains constant **OR** total momentum before (a collision) = total momentum after (a collision); [1 mark]
- Provided that there are no external / resultant forces acting; [1 mark]

(ii) Show that the mass of asteroid A was about $8.2 \times 10^{13} \text{ kg}$:

List the known quantities:

- Velocity of A, $m_A = 3.61 \text{ km s}^{-1} = 3.61 \times 10^3 \text{ m s}^{-1}$
- Momentum of A, $p_A = 2.00 \times 10^{17} \text{ kg m s}^{-1}$

Determine the mass of A:

- Momentum: $p = mv \Rightarrow m = \frac{p}{v}$
- $m_A = \frac{2.00 \times 10^{17}}{3.61 \times 10^3} = 5.5402 \times 10^{13} \text{ kg}$ [1 mark]
- Mass of asteroid A: $m_A = 5.54 \times 10^{13} \text{ kg}$ (3 or more s.f.) [1 mark]

[Total: 4 marks]

It's very common for students to miss the second marking point from their definitions of conservation of momentum – make sure you learn both parts!

Remember to use the unrounded calculated value for the mass of asteroid A

You should always work with more sig figs than you need during your calculation and then round at the end to avoid a cumulative rounding error

3b

(b) Calculate the combined velocity of the asteroids after the collision:

- Initial velocity of asteroid A, $u_A = 3.61 \times 10^3 \text{ m s}^{-1}$
- Momentum of asteroid A, $p_A = 2.00 \times 10^{17} \text{ kg m s}^{-1}$
- Mass of asteroid B, $m_B = 6.40 \times 10^{12} \text{ kg}$
- Initial velocity of asteroid B, $u_B = 14.0 \text{ km s}^{-1} = 14.0 \times 10^3 \text{ m s}^{-1}$

Use conservation of momentum:

- After the collision, asteroid B is absorbed by asteroid A, so the asteroids have a combined mass
- Therefore, $p_{\text{before}} = p_{\text{after}} \Rightarrow p_A + m_B u_B = (m_A + m_B) v$ [1 mark]

(b) Calculate the combined velocity of the asteroids after the collision:

- Initial velocity of asteroid A, $u_A = 3.61 \times 10^3 \text{ m s}^{-1}$
- Momentum of asteroid A, $p_A = 2.00 \times 10^{17} \text{ kg m s}^{-1}$
- Mass of asteroid B, $m_B = 6.40 \times 10^{12} \text{ kg}$
- Initial velocity of asteroid B, $u_B = 14.0 \text{ km s}^{-1} = 14.0 \times 10^3 \text{ m s}^{-1}$

Use conservation of momentum:

- After the collision, asteroid B is absorbed by asteroid A, so the asteroids have a combined mass
- Therefore, $p_{\text{before}} = p_{\text{after}} \Rightarrow p_A + m_B u_B = (m_A + m_B)v$ [1 mark]
- Making v the subject: $v = \frac{p_A + m_B u_B}{m_{AB}}$
- $v = \frac{(2.00 \times 10^{17}) + [(6.40 \times 10^{12}) \times (14.0 \times 10^3)]}{(5.54 \times 10^{13}) + (6.40 \times 10^{12})}$ [1 mark]
- $v = 4686 = 4.69 \text{ km s}^{-1}$ (3 s.f.) [1 mark]

[Total: 3 marks]

3c

(c) Show that the mass of asteroid C is about $1.0 \times 10^{13} \text{ kg}$:

List the known quantities:

- Mass of asteroid AB, $m_{AB} = (6.4 \times 10^{12}) + (5.54 \times 10^{13}) = 6.18 \times 10^{13} \text{ kg}$
- Mass of asteroid C = m_C
- Initial velocity of asteroid AB, $u_{AB} = 4.69 \text{ km s}^{-1}$
- Final velocity of asteroid AB, $v_{AB} = 1.2 \text{ km s}^{-1}$
- Final velocity of asteroid C, $v_C = 21.3 \text{ km s}^{-1}$
- Angle between AB and the horizontal, $\theta_{AB} = 8^\circ$

- Angle between C and the horizontal, $\theta_C = 10^\circ$

Calculate the initial momentum:

- Initial momentum: $p_{initial} = m_{AB} v_{AB}$
- $p_{initial} = (6.18 \times 10^{13}) \times (4.69 \times 10^3) = 2.896 \times 10^{17} \text{ kg m s}^{-1}$ [1 mark]

Calculate the final momentum:

- Final momentum: $p_{final} = m_{AB} v_{AB} \cos(\theta_{AB}) + m_C v_C \cos(\theta_C)$
- $p_{final} = [(6.18 \times 10^{13}) \times (1.2 \times 10^3) \times \cos 8^\circ] + [m_C \times (21.3 \times 10^3) \times \cos 10^\circ]$
- $p_{final} = (7.416 \times 10^{16}) \cos 8^\circ + m_C \times (21.3 \times 10^3) \cos 10^\circ$ [1 mark]

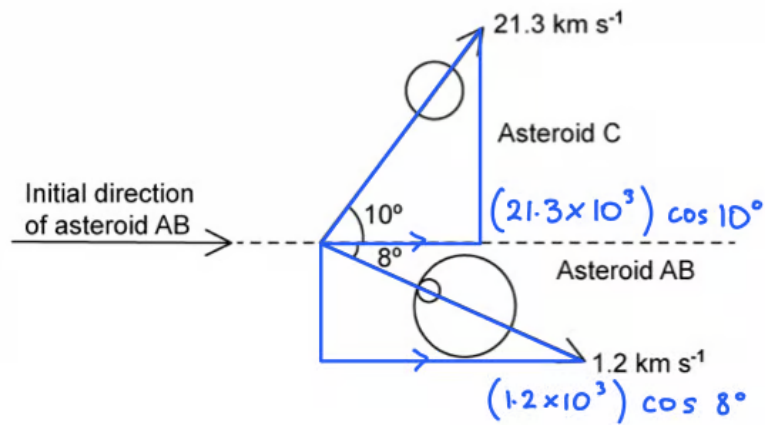
Use the conservation of momentum to calculate the mass of asteroid C:

- Total momentum before the collision = total momentum after the collision, or
 $p_{initial} = p_{final}$
- $2.896 \times 10^{17} = [(6.18 \times 10^{13}) \times (1.2 \times 10^3) \times \cos 8^\circ] + [m_C \times (21.3 \times 10^3) \times \cos 10^\circ]$
- $m_C = \frac{(2.896 \times 10^{17}) - (7.416 \times 10^{16}) \cos 8^\circ}{(21.3 \times 10^3) \cos 10^\circ}$ [1 mark]
- $m_C = 1.031 \times 10^{13} \text{ kg}$ [1 mark]

[Total: 4 marks]

Make sure you get lots of practice on momentum conservation in 2D scenarios, the extra step of resolving the components of vectors can slip a lot of students up!

We only need to consider the **horizontal components** of the velocities of the asteroids:



3d

(d) Determine if the collision in (c) is elastic or inelastic:

List the known quantities:

- Mass of asteroid AB, $m_{AB} = (6.4 \times 10^{12}) + (5.54 \times 10^{13}) = 6.18 \times 10^{13} \text{ kg}$
- Mass of asteroid C, $m_C = 1.031 \times 10^{13} \text{ kg}$
- Initial velocity of asteroid AB, $u_{AB} = 4.69 \text{ km s}^{-1}$
- Final velocity of asteroid AB, $v_{AB} = 1.2 \text{ km s}^{-1}$
- Final velocity of asteroid C, $v_C = 21.3 \text{ km s}^{-1}$

Calculate the initial kinetic energy:

- Initial kinetic energy: $KE_{initial} = \frac{1}{2} m_{AB} u_{AB}^2$
- $KE_{initial} = \frac{1}{2} \times (6.18 \times 10^{13}) \times (4.69 \times 10^3)^2$
- $KE_{initial} = 6.8 \times 10^{20} \text{ J}$ [1 mark]

Calculate the final kinetic energy:

- $KE_{final} = \frac{1}{2} m_{AB} v_{AB}^2 + \frac{1}{2} m_C v_C^2$
- $KE_{final} = \left[\frac{1}{2} \times (6.18 \times 10^{13}) \times (1.2 \times 10^3)^2 \right] + \left[\frac{1}{2} \times (1.031 \times 10^{13}) \times (21.3 \times 10^3)^2 \right]$

- $KE_{final} = 2.4 \times 10^{21} \text{ J}$ [1 mark]

Conclusion:

- An inelastic collision is one where kinetic energy is not conserved
- Since $KE_{initial} \neq KE_{final}$, this collision is inelastic; [1 mark]

[Total: 3 marks]

4a

(a) Determine the total change in momentum of the ball:

In an elastic collision:

- KE before = KE after
- Therefore, initial velocity = final velocity ($= 4.5 \text{ m s}^{-1}$) [1 mark]

The change in momentum is:

- Momentum before = mv
- Momentum after = $-mv$
- Change in momentum: $\Delta p = mv - (-mv) = 2mv$
- Change in momentum: $\Delta p = 2 \times (50 \times 10^{-3}) \times 4.5 = 0.45 \text{ kg m s}^{-1}$ [1 mark]

[Total: 2 marks]

4b

(b) Use the conservation of linear momentum to determine the rebound velocity of Y.

List the known quantities:

- Mass of X, $m_X = 50 \text{ g}$
- Mass of Y, $m_Y = 125 \text{ g}$
- Initial velocity of X, $u_X = 4.5 \text{ m s}^{-1}$
- Final velocity of X, $v_X = -2.8 \text{ m s}^{-1}$
- Initial velocity of Y, $u_Y = -2.2 \text{ m s}^{-1}$

- Final velocity of Y, $v_Y = v$

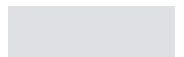
The momenta before and after the collision:

- Momentum before = $m_X u_X + m_Y u_Y$
- Momentum before = $(50 \times 4.5) + (125 \times -2.2)$ [1 mark]
- Momentum after = $m_X v_X + m_Y v_Y$
- Momentum after = $(50 \times -2.8) + (125 \times v)$ [1 mark]

Conservation of linear momentum:

- Momentum before = momentum after: $m_X u_X + m_Y u_Y = m_X v_X + m_Y v_Y$
- $(50 \times 4.5) + (125 \times -2.2) = (50 \times -2.8) + (125 \times v)$
- $225 - 275 = -140 + 125v \Rightarrow 125v = 90$
- Rebound velocity of Y: $v = \frac{90}{125} = 0.72 \text{ m s}^{-1}$ [1 mark]

[Total: 3 marks]



The momenta before and after the collision:

- Momentum before = $m_X u_X + m_Y u_Y$
- Momentum before = $(50 \times 4.5) + (125 \times -2.2)$ [1 mark]
- Momentum after = $m_X v_X + m_Y v_Y$
- Momentum after = $(50 \times -2.8) + (125 \times v)$ [1 mark]

Conservation of linear momentum:

- Momentum before = momentum after: $m_X u_X + m_Y u_Y = m_X v_X + m_Y v_Y$
- $(50 \times 4.5) + (125 \times -2.2) = (50 \times -2.8) + (125 \times v)$
- $225 - 275 = -140 + 125v \Rightarrow 125v = 90$
- Rebound velocity of Y: $v = \frac{90}{125} = 0.72 \text{ m s}^{-1}$ [1 mark]

[Total: 3 marks]

In this question, you do not need to convert the mass of the balls to kg. Both balls have their mass in grams and when divided in g or kg the ratio will be the same.

4c

(c) Deduce whether the collision is elastic or inelastic

List the known quantities:

- Initial velocity of X, $u_X = 4.5 \text{ m s}^{-1}$
- Final velocity of X, $v_X = -2.8 \text{ m s}^{-1}$
- Initial velocity of Y, $u_Y = -2.2 \text{ m s}^{-1}$
- Final velocity of Y, $v_Y = 0.5 \text{ m s}^{-1}$

An elastic collision is...

- A collision in which kinetic energy is conserved **or** relative speed of approach =

relative speed of separation

Write a conclusion:

- Calculation showing that total initial KE $\neq$ the total final KE **or** relative speed of approach $\neq$ relative speed of separation; [1 mark]

e.g. the relative speeds of approach and separation are:

- Relative speed of approach: $u_X - u_Y = 4.5 - (-2.2) = 6.7 \text{ m s}^{-1}$
 - Relative speed of separation: $v_Y - v_X = 0.5 - (-2.8) = 3.3 \text{ m s}^{-1}$
 - The relative speed of approach (6.7 m s^{-1}) $\neq$ relative speed of separation (3.3 m s^{-1})
- Therefore, the collision is not elastic **or** the collision is inelastic; [1 mark]

[Total: 2 marks]

4d

(d) Use Newton's laws to explain why the magnitude of the change in momentum of each ball is the same

Newton's third law states...

- When two objects interact, they exert equal and opposite forces on each other
- Therefore, the force on X is equal (in magnitude) and opposite (in direction) to the force on Y; [1 mark]

Newton's second law states...

- The resultant force on an object is equal to the rate of change of momentum (i.e. $\Delta p / \Delta t$) of the object, and it acts in the same direction, or $F = \Delta p / \Delta t$
- Therefore, the force (on X & Y) is equal to / proportional to the rate of change of momentum; [1 mark]

The change in momentum of X & Y is the same because...

- The time of collision is the same for both balls (hence change in momentum is the same); [1 mark]

[Total: 3 marks]

5a

(a) The total momentum of the rifle and the bullet before being fired is:

- Zero / 0 kg m s^{-1} [1 mark]

Because:

- (Before the rifle is fired both) rifle and bullet are stationary / velocity = 0; [1 mark]

[Total: 2 marks]

5b

(b) Calculate the velocity of the shell just after the rifle is fired

List the known quantities:

- Mass of shell, $m_s = 30 \text{ g} = 0.03 \text{ kg}$
- Momentum of shell after being fired, $p = 36 \text{ kg m s}^{-1}$

Calculate the velocity of the shell using the momentum equation:

- Momentum: $p = mv$
- $v = \frac{p}{m} = \frac{36}{0.03}$ [1 mark]
- $v = 1200 \text{ m s}^{-1}$ [1 mark]

[Total: 2 marks]

For the velocity to be in m s^{-1} , the momentum must be in kg m s^{-1} and the mass in **kg** – unit conversions are very common, don't let these catch you out!

5c

(c)

(i) The total momentum of the rifle and the shell after the rifle is fired is:

- Total momentum before the rifle is fired = total momentum after the rifle is fired **OR**
 $p_{initial} + p_{final} = 0$
- Total momentum = 0 kg m s⁻¹ [1 mark]

(ii) Calculate the recoil momentum of the rifle:

List the known quantities:

- Total final momentum of the rifle and shell, $p_f = 0$
- Momentum of the shell after being fired, $p_s = 36 \text{ kg ms}^{-1}$ (from **(b)**)

Decide on the positive direction:

- Since $p_s = 36 \text{ kg m s}^{-1}$ then the motion of the shell gives the positive direction
- Therefore the recoil will have a negative value **OR** $p_s = -p_r$

Calculate:

- Therefore $p_s = -p_r = 36$ [1 mark]
- $p_r = -36 \text{ kg m s}^{-1}$ [1 mark]

[Total: 3 marks]

You may have chosen the recoil of the rifle to represent the **positive** direction. In this case, your values should be the same as ours, but the negative and positive signs will be reversed.

5d

(d) Calculate the average force needed to stop the bullet

List the known quantities:

- Momentum of shell before hitting target, $p_s = 9.8 \text{ kg m s}^{-1}$
- Time taken for shell to stop, $\Delta t = 4.5 \text{ ms} = 4.5 \times 10^{-3} \text{ s}$

Determine the change in momentum Δp as the bullet is stopped by target:

- The shell comes to rest, therefore final momentum = 0
- Change in momentum: $\Delta p = 9.8 \text{ kg m s}^{-1}$ [1 mark]

Calculate the average force F needed to stop the bullet:

- Force & momentum: $F = \frac{\Delta p}{\Delta t}$
- Average force: $F = \frac{9.8}{4.5 \times 10^{-3}}$ [1 mark]
- Average force: $F = 2177.8 = 2200 \text{ N (2 s.f.)}$ [1 mark]

[Total: 3 marks]

Calculations like this one are usually given what examiners call 'error carried forward' (you will see it written 'ecf' in mark schemes).

This means that, for example, if you forgot to convert ms to s (using 4.5 s rather than 0.0045) then you could still get the other three marks, as long as your final answer was 2200 N.

However, if you did get this then your 'sanity check' at the end should have seen you looking at this number and realising it must be wrong, since it suggests that being shot by a rifle feels similar to having someone pat you lightly on the back!

